ImmunoBlitz: 50-Minute Science Lesson Plan

Using gameplay to explore infection, immunity and scientific models

AGE	DURATION	SUBJECT	TOPIC
11-13	50 minutes	Science	Infection & immunity

Lesson Purpose

Students use ImmunoBlitz as an interactive model of infection and immunity. The lesson combines gameplay, observation, discussion and evidence checking so that students learn both biological ideas and the limitations of a simplified scientific model.

Learning Objectives

- Identify examples of pathogens, body defences and health interventions represented in ImmunoBlitz.
- Explain that different threats require different responses, for example antibiotics act against bacteria rather than viruses.
- Distinguish between a game mechanic and the real biological process it represents.
- Evaluate one way ImmunoBlitz simplifies real biology and explain why the simplification may still be useful for learning.

Success Criteria

By the end of the lesson, students should be able to describe at least one pathogen or defence from the game, explain one science-based gameplay decision, and identify one difference between ImmunoBlitz and real biology.

Materials & Preparation

- A computer or tablet with access to ImmunoBlitz for each student or pair.
- ImmunoBlitz Gameplay Investigation Worksheet (Step 3A).
- ImmunoBlitz Game vs. Real Biology Activity (Step 3B) - use selected questions during this lesson or as follow-up.
- Access to the ImmunoBlitz Learning Hub or a teacher-selected Learning Hub article.
- Board/projector for the starter and class discussion.

Lesson Sequence

Time	Stage	Student activity	Teacher focus
0-5 min	**Starter**	Ask: What stops pathogens from making us ill every time we encounter them? Collect ideas such as skin, mucus, white blood cells, antibodies, vaccines, antibiotics and hygiene.	Keep answers visible. Do not correct every misconception yet; return to the list later.
5-8 min	**Set the task**	Explain that students will play ImmunoBlitz as scientists, not only as players. Give out the Gameplay Investigation Worksheet and briefly model how to record a decision.	Emphasise: record what you chose, what happened, and whether the choice was based on biology, game strategy, or both.
8-23 min	**Gameplay**	Students play ImmunoBlitz and complete the observation table. They should record at least three decisions involving a pathogen, defence or intervention.	Circulate and ask probing questions: Why did you choose that? What real process might this represent?
23-33 min	**Compare & discuss**	In pairs, students compare one decision and identify one game mechanic that is scientifically useful but simplified. Use a selected prompt from the Game vs. Real Biology Activity.	Bring out examples such as waves, health bars, instant upgrades, visible pathogens or simplified intervention effects.
33-45 min	**Evidence check**	Students use a relevant Learning Hub article to check one claim or mechanic from the game. They note what the article confirms, adds or corrects.	Suggested articles: pathogen differences, first-line defences, cellular defenders, vaccines, antibiotics, or the modelling article.
45-50 min	**Exit ticket**	Students answer: 1) One thing ImmunoBlitz represents well. 2) One thing it simplifies. 3) One science idea I understand better after playing.	Collect responses to identify misconceptions and decide what needs revisiting next lesson.

Assessment

- Formative: listen for explanations during gameplay and pair discussion rather than only checking whether students can name terms.
- Worksheet evidence: students should connect a gameplay decision to a biological idea.
- Exit ticket: use responses to identify misconceptions, especially 'all microbes are harmful', 'antibiotics treat viruses', or 'the game is a literal simulation'.

Teacher Notes, Differentiation & Follow-Up

Key Discussion Points

- Pathogens are not one single type of organism: viruses, bacteria and fungi differ biologically.
- The body uses several overlapping layers of defence, including barriers and immune cells.
- Prevention and treatment are not the same thing. Vaccines, antibiotics, hygiene, masks and other interventions have different purposes.
- A game mechanic can be educationally useful even when it is not literally realistic.
- ImmunoBlitz should be used as a starting point for questioning and investigation, not as a medical simulation.

Differentiation

Support

Pair students, provide key vocabulary, and use sentence starters such as: "The game represents ___ by ___" and "This is simplified because in real life ___."

Challenge

Ask students to identify a possible misconception caused by a mechanic, verify it using a Learning Hub source, and propose a scientifically clearer mechanic that would still work as a game.

Common Misconceptions to Watch For

- All microbes are harmful.
- Antibiotics kill viruses.
- The immune system is only white blood cells.
- Vaccination works instantly.
- Real infections happen in neat waves or organs have a single 'health percentage'.
- Every game choice represents something a person can consciously control in the real body.

Optional Follow-Up

- Complete the full Game vs. Real Biology Activity as homework or in the next lesson.
- Ask students to design one new ImmunoBlitz mechanic and justify the science behind it.
- Use a different Learning Hub article for a short evidence-checking or source-evaluation task.
- Revisit the starter list and ask students to correct or reorganise their original ideas.

Suggested Learning Cycle

Play -> Notice -> Question -> Investigate. The game creates the experience, classroom discussion exposes the model, and the Learning Hub provides the deeper scientific explanation.